

Green's Theorem in the Plane.

→ If $M(x, y)$ and $N(x, y)$ be continuous functions of x and y having continuous partial derivatives $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ in a region R of the xy -plane bdd by a closed curve C , then

$$\oint_C (M dx + N dy) = \iint_R \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dx dy$$

where C is traversed in the counterclockwise direction.

6. Verify Green's theorem in the plane for $\oint_C (2x^2 - 8y^2) dx + (4y - 6xy) dy$ where C is the boundary of the region defined by,

a) $y = \sqrt{x}$, $y = x^2$ b) $x = 0$, $y = 0$, $x + y = 1$.

A) $y = \sqrt{x}$, $y = x^2$, $M = 2x^2 - 8y^2$, $N = 4y - 6xy$

$$\frac{\partial M}{\partial y} = -16y$$

$$\frac{\partial N}{\partial x} = -6y$$

$$\therefore \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 10y$$

If R is the region bdd by C , then

$$\iint_R \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dx dy$$

$$= \int_0^1 \int_{x^2}^{\sqrt{x}} 10y dy dx$$

$$= 5 \int_0^1 [y^2]_{x^2}^{\sqrt{x}} dx$$

$$= 5 \int_0^1 (x - x^4) dx$$

$$= 5 \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1$$

$$= 5 \left[\frac{1}{2} - \frac{1}{5} \right]$$

$$= 5 \times \frac{3}{10} = \frac{3}{2} \quad \text{--- ①}$$

$$\text{Also } \oint_C M dx + N dy = \int_{C_1} (M dx + N dy) + \int_{C_2} M dx + N dy \quad \text{--- ②}$$

Along C_1 , $x^2 = y \Rightarrow 2x dx = dy$, $0 \leq x \leq 1$

$$\begin{aligned} \therefore \int_{C_1} M dx + N dy &= \int_0^1 (x^2 - 8x^4) dx + (4x^2 - 6x \cdot x^2) 2x dx \\ &= \int_0^1 (3x^2 + 8x^3 - 20x^4) dx \\ &= \left[x^3 + 2x^4 - 4x^5 \right]_0^1 = 1 + 2 - 4 = \underline{\underline{-1}} \end{aligned}$$

Along C_2 , $y^2 = x \Rightarrow 2y dy = dx$, $0 \leq y \leq 1$

$$\begin{aligned} \therefore \int_{C_2} M dx + N dy &= \int_0^1 (3y^4 - 8y^2) 2y dy + (4y - 6y^2 \cdot y) dy \\ &= \int_0^1 (4y - 22y^3 + 6y^5) dy = \left[2y^2 - \frac{11}{2} y^4 + y^6 \right]_0^1 \\ &= \left(0 - 2 + \frac{11}{2} - 1 \right) = \underline{\underline{5/2}} \end{aligned}$$

$$\therefore \text{②} \Rightarrow \oint_C M dx + N dy = -1 + 5/2 = 3/2 \quad \text{--- ③}$$

$\therefore$ equality of ① and ③ verifies Green's thm in the plane.

$$\begin{aligned} \text{b) Here } \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \int_0^1 \int_0^{1-x} 10xy dy dx \\ &= \int_0^1 5y^2 \Big|_0^{1-x} dx \\ &= 5 \int_0^1 (1-x)^2 dx \end{aligned}$$

$$= 5 \left[\frac{(1-x)^3}{-3} \right]_0^1$$

$$= -5/3 [0-1] = \underline{\underline{5/3}} \text{ --- ①}$$

Along OA, $y=0$, $dy=0$, $0 \leq x \leq 1$.

$$\int_{OA} M dx + N dy = \int_0^1 3x^2 dx$$

$$= x^3 \Big|_0^1 = \underline{\underline{1}}$$

Along AB, $y=1-x$, $dy=-dx$, $1 \leq x \leq 0$

$$\int_{AB} M dx + N dy = \int_1^0 [3x^2 - 8(1-x^2)] dx + [4(1-x) - 6x(1-x)] (-dx)$$

$$= \int_1^0 (3x^2 - 8 + 8x^2 - 4 + 4x + 6x - 6x^2) dx$$

$$= \int_1^0 (-12 + 26x - 11x^2) dx$$

$$= \left[-12x + 13x^2 - 11x^3/3 \right]_1^0$$

$$= [0 - (-12 + 13 - 11/3)]$$

$$= \underline{\underline{8/3}}$$

Along BO, $x=0$, $dx=0$, $1 \leq y \leq 0$.

$$\int_{BO} M dx + N dy = \int_1^0 4y dy = 2y^2 \Big|_1^0 = -2.$$

$$\therefore \int_C M dx + N dy = 1 + 8/3 - 2 = 5/3 \text{ --- ②} \quad \begin{matrix} \text{①} = \text{②} \\ \Rightarrow \text{C.T} \end{matrix}$$

Q. Use Green's thm in a plane to evaluate the integral $\oint_C (2x^2 - y^2) dx + (x^2 + y^2) dy$, where C is the boundary in the xy -plane of the area enclosed by the x -axis and the semi-circle $x^2 + y^2 = 1$ in the upper half xy -plane?

If R is the region bounded by the closed curve C , then by Green's thm,

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy.$$

$$M = 2x^2 - y^2, \quad N = x^2 + y^2$$

$$\frac{\partial M}{\partial y} = -2y, \quad \frac{\partial N}{\partial x} = 2x.$$

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 2x + 2y$$

$$= 2(x + y).$$

R is bdd by

$$x = -1, x = 1, y = 0, y = \sqrt{1 - x^2}.$$

$$\therefore \oint_C (2x^2 - y^2) dx + (x^2 + y^2) dy = \iint_R 2(x + y) dx dy$$

$$= 2 \int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x + y) dy dx$$

$$= 2 \int_{-1}^1 \left(xy + \frac{y^2}{2} \right) \Big|_0^{\sqrt{1-x^2}} dx$$

$$= 2 \int_{-1}^1 (x\sqrt{1-x^2} + 1 - x^2) dx$$

$$\begin{aligned}
 &= 2 \left[\int_{-1}^1 x \sqrt{1-x^2} dx + \int_{-1}^1 1-x^2 dx \right] \quad \begin{aligned} 1-x^2 &= u \\ -2x dx &= du \end{aligned} \\
 &= 2 \left[0 + \left[x - \frac{x^3}{3} \right]_{-1}^1 \right] \quad \begin{aligned} x dx &= -\frac{1}{2} du \\ x=1, u=0 \\ x=-1, u=0 \end{aligned} \\
 &= 2 \left[1 - \frac{1}{3} + 1 - \frac{1}{3} \right] \\
 &= 2 \left[2 - \frac{2}{3} \right] = 2 \times \frac{(6-2)}{3} = \underline{\underline{\frac{8}{3}}}
 \end{aligned}$$

Stoke's Theorem (Relation b/w line and surface integrals)

→ Let S be an open surface bdd by a closed curve C and $\vec{F} = F_1 i + F_2 j + F_3 k$ be any vector pt fn having contin first order partial derivatives. Then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, dS, \text{ where } \hat{n} \text{ is a unit}$$

vector at any point of S drawn in the sense in which

Q. Verify Stoke's thm for $\vec{F} = (x^2+y^2)i - 2xyj$ taken around the rectangle bdd by the lines $x=\pm a, y=0, y=b$

Let C denote the bdr of the rectangle ABED. Then

$$\begin{aligned}
 \oint_C \vec{F} \cdot d\vec{r} &= \oint_C (x^2+y^2)i - (2xy)j \cdot (x dx + y dy) \\
 &= \oint_C (x^2+y^2) dx - 2xy dy
 \end{aligned}$$